

ADK: Building AI Agents

with Google's Agent Development Kit

A BEGINNER'S GUIDE FOR COLLEGE STUDENTS

ADK is an open-source toolkit for building AI agents

- Agent Development Kit, created by Google
- Helps you build programs that reason, use tools, and complete tasks
- Works with code, mainly Python and other languages
- Supports single agents and teams of agents working together

AI Agent Workflow

Reasoning

Analyze goals and context

Use Search

Find external information

Pick Tool

Select Search, Database, Calendar

Use Database

Retrieve structured records

Where students and teams use ADK



Study assistants

Search notes and answer questions



Customer support bots

Look up information and act on it



Research helpers

Gather and summarize sources



Workflow automation

Scheduling or report writing



Advantages and disadvantages

Advantages

- Free and open source, so easy to learn with
- Modular design: build agents from small reusable parts
- Flexible: works with different AI models and tools

Disadvantages

- Requires basic programming knowledge
- Still new, so features and docs change quickly
- Agents can make mistakes and need testing and oversight

Conclusion: a practical first step into AI agents

- 01 ADK turns AI ideas into working, testable agents
- 02 Strong for learning, projects, and real applications
- 03 Next step: build a small agent for your own coursework

